

1. Executive Summary & Architecture

StockFlow is an enterprise inventory management and shift auditing system engineered for high-accuracy medical and commercial environments. It automates inventory tracking, shift handovers, financial valuation, and security auditing.

I SYSTEM ARCHITECTURE & TECH STACK

FRONTEND ARCHITECTURE

Built with **Vanilla HTML5, ES6+ JavaScript, and CSS3**. Operates dependency-free for maximum browser performance, featuring responsive Tailwind-styled components, modal dialogues, and live client-side data filters.

BACKEND REST API

Powered by **Node.js & Express.js v5**. Implements stateless JWT authentication, password hashing with `bcryptjs`, parameterized SQL protection, and structured JSON endpoint handlers.

I CORE COMPONENT SCHEMA

MODULE	TECHNOLOGY	KEY RESPONSIBILITY	SECURITY CONTROL
Auth Engine	JWT + Bcrypt	User login, token issuance, password validation	RBAC ADMIN GUARD
Inventory Module	Express + SQL	Stock entries, opening/closing calculations, shift linking	OWNERSHIP CHECK
Financial Engine	PostgreSQL CTE	Stock valuation (In, Out, Damaged, Current)	ADMIN ONLY
PDF Renderer	Offscreen Iframe	Compiled client-side PDF document generation	ISOLATED CONTEXT

I SECURITY & ROLE-BASED ACCESS CONTROL (RBAC)

STANDARD STAFF USER

- Record daily morning and night shift stock counts.
- View active product catalogue and self-created entries.
- Restricted from modifying entries created by other staff.

SYSTEM ADMINISTRATOR

- Full CRUD permissions across Products, Users, and Stock.
- Access to Audit Logs, Financial Valuations, & Loss Reports.
- Database auto-recovery and system diagnostics rights.

2. Shift Operations & User Guide

StockFlow streamlines daily inventory workflows by automatically bridging morning closing stock with night shift opening stock, preventing unrecorded inventory leakage.

MANDATORY SHIFT HANDOVER LOGIC

Expected Stock = Opening + Received - Stock Out - Damaged
Variance = Physical Closing Count - Expected Stock

```
// Automated Shift Handover Algorithm (inventoryController.js)
if (shift === 'night') {
  const morning = await getMorningClosing(product_id, date);
  if (morning) {
    openingStock = morning.closing; // Auto-link Night Opening to Morning Closing
    variance = physicalClosing - (openingStock + received - damaged - disbursed);
  }
}
```

DAILY OPERATIONAL STEPS

1. AUTHENTICATION & SHIFT SELECTION

Staff log in using their assigned credentials. Upon initiating entries, the session shift (**Morning** or **Night**) and active working date are established.

2. RECORDING PRODUCT STOCK

Staff input physical counts for **Received**, **Disbursed (Stock Out)**, **Damaged**, and **Closing Count**. The system instantly computes variances and alerts staff if physical counts deviate from calculated totals.

3. CATALOGUE MANAGEMENT (ADMIN)

Admins manage items via the Products Catalogue module. Product names are automatically normalized (e.g., `lower(name)` deduplication) to prevent duplicate inventory entries.

INTERFACE DESIGN & RESPONSIVENESS

- Desktop View:** High-density data grid with position: sticky header rows for fast scrolling during bulk shift audits.
- Mobile View:** Touch-optimized card list with bounded containers (`max-h-[60vh] overflow-y-auto`) ensuring responsive layout on smartphone screens.

3. Financial Reporting & Maintenance

StockFlow provides real-time financial reporting for management alongside an append-only audit trail and automated database disaster recovery options.

I FINANCIAL STOCK VALUE SUMMARY

TOTAL RECEIVED Received Value	TOTAL EXITED Stock Out Value	DAMAGED LOSS Financial Loss	VALUATION Current Value
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I ISOLATED PDF GENERATION ENGINE

PDF reports (Shift Reports, Audit Logs, Financial Analytics) use an isolated offscreen rendering engine (exportHTMLToPDF) to guarantee zero canvas clipping or blank pages:

- **Offscreen Iframe Isolation:** Mounts content in a hidden iframe (1040px landscape / 740px portrait) to bypass parent CSS rules.
- **Base64 Asset Embedding:** Injects the high-resolution **DMC Hospital Logo** directly via inline Base64 data strings.
- **Instant Preview & Download:** Generates a downloadable .pdf file and opens a viewable browser blob tab (output('bloburl')).

I AUDIT TRAIL & DATABASE AUTO-RECOVERY

All system actions (CREATE, UPDATE, DELETE) write to an immutable audit_logs table containing previous and updated JSON snapshots. Setting AUTO_RESTORE=true enables automated disaster recovery from audit logs.

I ENVIRONMENT SETUP GUIDE

```
# backend/.env Configuration
PORT=5000
NODE_ENV=production
DATABASE_URL=postgresql://user:pass@host:5432/stockflow
JWT_SECRET=your_super_secret_jwt_key
JWT_EXPIRES_IN=8h
AUTO_RESTORE=false
```